American Computer Science League

2021-2022 ● Contest 1: Fibonacci Clock ● Junior Division

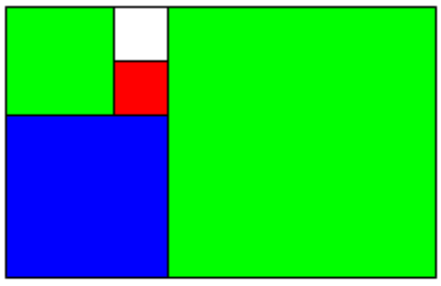


PROBLEM: ACSL's version of Philippe Chretien's "Fibonacci Clock" displays time by changing the colors displayed in 5 squares, whose side lengths correspond to the first 5 Fibonacci numbers (1, 1, 2, 3, and 5). Given the colors of the squares on the clock face, you must output the time that is represented in hh:mm format. The colors will be given to you as 5 individual characters representing the lower 1x1 square, the upper 1x1 square, the 2x2 square, the 3x3 square, and the 5x5 square in that order. (See https://basbrun.com/2015/05/04/fibonacci-clock/.)

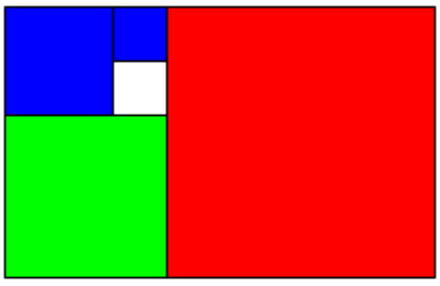
Red squares are used to represent only hours and green squares are used to represent only minutes. Blue squares are used to represent both hours and minutes. White squares are ignored. To find the current hour, add the values of the red and blue squares. To find the current minutes, add the values of the green and blue squares and multiply by 5. The minutes are displayed in five-minute increments.

The clock in the picture above is displaying 08:50. The hours are represented by the red 1x1, blue 2x2, and blue 5x5 squares (1+2+5=8). The minutes are represented by the blue 2x2, the green 3x3, and the blue 5x5 squares (2+3+5=10 and 10*5=50).

The example at the left which uses the inputted characters R, W, G, B, G displays the time 04:50. The hours are represented by the 1x1 red square and the 3x3 blue square: 1+3=4. The minutes are represented by the 2x2 green, the 3x3 blue, and the 5x5 green squares: (2+3+5)*5 = 50. The example at the right which uses the inputted characters W, B, B, G, R displays the time 08:30. The hours are 1+2+5=8 and the minutes are (1+2+3)*5=30.



R W G B G - 04:50



W B B G R - 08:30

American Computer Science League

2021-2022 ● Contest 1: Fibonacci Clock ● Junior Division

INPUT: There are 5 sets of data. Each set has 5 uppercase letters (R, G, B, or W) that represent the colors of the lower 1x1, the upper 1x1, the 2x2, the 3x3, and finally the 5x5 square, in that order. We guarantee that the input will represent a valid time from 00:00 to 11:55.

OUTPUT: For each line of data, print the time in hours and minutes formatted as hh:mm.

SAMPLE INPUT:

```
R W G B G
W B B G R
B G B B R
W W W B B
W R G G G
```

SAMPLE OUTPUT:

```
1. 04:50
2. 08:30
3. 11:35
4. 08:40
5. 01:50
```

American Computer Science League

2021-2022 ● Contest 1: Fibonacci Clock ● Junior Division

TEST DATA

TEST INPUT:

```
G R W B B
R B B W G
W W W W W
B W W G R
W B B B B
```

TEST OUTPUT:

1. 09:45
2. 04:40
3. 00:00
4. 06:20
5. 11:55

美国计算机科学联赛

2021-2022 ● 第一轮: 斐波那契时钟 ● 初级组

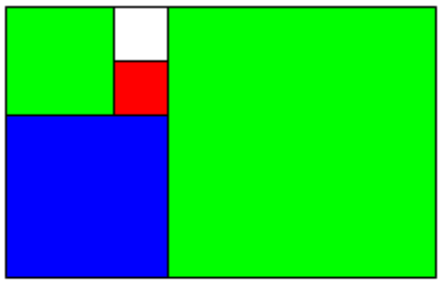


问题：美国计算机科学联赛版本的菲利普·克雷蒂安“斐波那契时钟”是通过改变由五**个**正方形**呈现**的颜色来显示时间的，它们的边长分别对应斐波那契数**列**的前五个数（1，1，2，3，5）。你要根据钟面上正方形的颜色，以 hh:mm 的格式输出其对应的时间。这些颜色会以**五个单独**字母的方式给出，依次代表下方 1×1 正方形，上方 1×1 正形，2×2 正方形，3×3 正方形，以及 5×5 正方形。(见 https://basbrun.com/2015/05/04/fibonacci-clock/.)

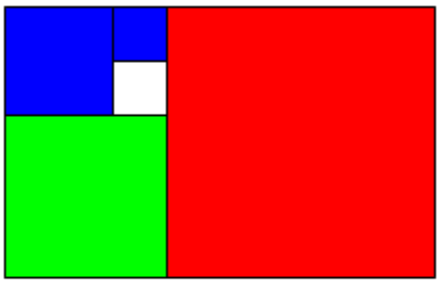
红色正方形只能表示小时，绿色正方形只能表示分钟。蓝色正方形既可以表示小时，也可以表示分钟。白色正方形**可以**忽略。将红、蓝色正方形对应的值相加得出当前的小时，将绿、蓝色正方形的值相加再乘以 5 得出当前的分钟。**分钟数以 5 为间隔**。

上图的时钟显示的是时间是08:50。1x1的红色正方形， 2x2的蓝色正方形以及5x5的蓝色正方形 (1+2+5=8) 表示小时。2x2的蓝色正方形， 3x3的绿色正方形以及5x5的蓝色正方形表示分钟数 (2+3+5=10, 且 10*5=50)。

下方左边的示例通过输入字母 R, W, G, B, G 来显示时间 04:50。用1×1 的红色正方形和 3x3 的蓝色正方形来表示小时数：1+3=4。用 2×2 的绿色正方形，3×3的蓝色正方形以及5×5的绿色正方形来表示分钟数： (2+3+5)*5 = 50。右边的示例通过输入字母 W, B, B, G, R 来显示时间 08:30。小时数为：1+2+5=8 ，分钟数为 (1+2+3)*5=30。



R W G B G - 04:50



W B B G R - 08:30

美国计算机科学联赛

2021-2022 ● 第一轮: 斐波那契时钟 ● 初级组

输入：有五组数据。每组包含 5 个大写字母 (R, G, B, 或 W)，这些字母依次表示下方 1x1 正方形, 上方1×1正方形, 2×2 正方形, 3×3 正方形以及最后的 5×5 正方形。确保输入内容能表示 00:00 到 11:55 之间的一个有效时间。

输出：打印输出每一行数据中由小时和分钟组成的时间，格式为 hh:mm。

示例输入：	示例输出：
R W G B G	1. 04:50
W B B G R	2. 08:30
B G B B R	3. 11:35
W W W B B	4. 08:40
W R G G G	5. 01:50

美国计算机科学联赛

2021-2022 ● **第一轮**: 斐波那契时钟 ● 初级组

测试数据

测试输入:

```
G R W B B
R B B W G
W W W W W
B W W G R
W B B B B
```

测试输出:

1. 09:45
2. 04:40
3. 00:00
4. 06:20
5. 11:55